A Computational View of Medicine

Why AI will matter most where disease becomes a search problem

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March 17, 2026

Thesis. The most important medical use of advanced AI is not generic automation. It is high-context reasoning over difficult, patient-specific problems where better synthesis can change diagnosis, treatment choice, escalation, or trial matching. In these settings, disease is better understood as a computational burden distributed across time, modalities, and decision branches than as a static label alone.

Abstract

Artificial intelligence will not improve every part of medicine equally. Its highest-value role is emerging in disease settings where the central bottleneck is not data collection alone or workflow automation alone, but high-context reasoning over multimodal evidence under uncertainty. We argue for a computational view of medicine: one that treats many serious illnesses as search problems over longitudinal records, pathology, imaging, genomics, patient-generated data, prior treatment history, and branching therapeutic options. Under this view, the key design question is not whether a model can produce clinically fluent text, but whether additional inference changes the next real decision for a specific patient. We synthesize evidence from precision oncology, rare-disease genomics, antimicrobial discovery, sepsis surveillance, patient-reported outcome monitoring, open-notes adoption, and emerging N-of-1 medicine. We further argue that patient access to records is necessary but insufficient; the practical value of data access rises sharply when patients and clinicians can repeatedly compute over the full timeline of illness. The result is a framework for directing AI toward cases where reasoning is the bottleneck, outcomes are sensitive to synthesis quality, and patient-held data can become a force multiplier rather than a static archive.

1 Introduction

Medicine has long been organized around disease labels, organ systems, and specialty workflows. Those abstractions are useful, but they often obscure where the real difficulty lies. In many of the cases that matter most, the limiting factor is neither the absence of a diagnosis code nor the absence of a single decisive test. It is the burden of synthesis: reconciling longitudinal evidence, deciding which signals matter, ranking plausible interventions, and updating the model of the case as new data arrive.

This paper argues that advanced AI should be judged through that lens first. The central question is not whether a system can imitate clinical language. It is whether additional inference changes a meaningful next decision in settings where the decision is difficult, high-stakes, and patient-specific. This framing is consistent with the broader case for human-plus-machine medicine made by Topol, but it places computational leverage rather than generic augmentation at the center [29].

Crucially, this computational constraint is not unique to machines. It is already the daily reality of clinical practice. Physicians employ both rapid pattern recognition and slower deliberate analysis—what cognitive science calls dual-process reasoning—and they switch between these modes depending on case complexity [7]. But the time a clinician can spend thinking about any single patient is finite. If a doctor had twice as long to reason through a complex case, would they reach a better conclusion? In many routine situations, the answer is no: the protocol is clear and more thought adds nothing. But in high-burden cases—a confusing constellation of symptoms, an ambiguous genomic result, a cancer that has stopped responding—more reasoning time genuinely changes the output. This is not merely a theoretical claim. The "diagnostic timeout" in emergency medicine—a deliberate, structured pause to reconsider the differential diagnosis—has been shown to reduce diagnostic error, providing direct empirical evidence that allocating more cognitive compute to a case works in practice [7, 9]. Furthermore, primary care consultation lengths vary sharply across international healthcare systems, ranging from 5 minutes in high-volume settings to more than 20 minutes in others [13]. If reasoning time is the central bottleneck the paper claims it is, this variation should predict diagnostic quality, grounding the computational metaphor directly in healthcare economics.

Recent work on reasoning-focused language models suggests that AI systems can engage in structured clinical reasoning that parallels this deliberate process, including hypothesis generation, differential weighting, and evidence integration. However, the relationship between reasoning length and decision quality is not monotonic. Evaluations of models like DeepSeek R1 reveal a clear tension: shorter reasoning chains often correlate with accuracy, while excessively long chains correlate with errors driven by anchoring bias, overthinking, and rationalizing wrong conclusions [20]. The value of additional reasoning depends heavily on its functional form. It is productive when it widens the search to explore new hypothesis space, but it becomes destructive when it merely deepens commitment to a single, incorrect branch. This distinction separates good clinical reasoning from bad clinical reasoning in humans, and it turns "more reasoning helps" from a slogan into a concrete design principle for medical AI. The implication is not that machines reason exactly as clinicians do, but that diagnosis and treatment selection are search problems, and the quality of the search depends on the structure, not just the duration, of the reasoning applied.

The argument has three parts. First, disease classes differ sharply in how much more computation can improve outcomes. Second, patient access to longitudinal records changes the practical location of medical intelligence: once records can be assembled across sites, compute can move with the patient rather than remaining trapped inside an institution. Third, the future of high-value health AI is likely to look increasingly N-of-1: iterative, longitudinal, individualized, and explicitly tied to decisions rather than one-shot answers.

Definition. A *computational view of medicine* treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty.

2 Scope and evidence selection

This paper is a scientific perspective rather than a systematic review. The evidence base was assembled to support a specific claim: that the clinical value of AI is concentrated in disease settings where the main bottleneck is patient-specific synthesis under uncertainty. We therefore prioritized five categories of evidence: prospective interventional studies in which computational systems changed care operations or outcomes; landmark mechanistic or translational studies that make the search-space argument concrete; studies on patient access to records and patient-generated data; literature on N-of-1 and individualized therapeutic design; and implementation-oriented work on digital phenotyping, quality, and safety [12, 15, 18].

The paper does not claim that all cited domains already have mature evidence for the full computational paradigm. Instead, it distinguishes three layers of support: direct interventional evidence that computation changed outcomes; translational evidence that a disease domain is intrinsically computational; and forward-looking inference about how patient-held data and repeated synthesis could alter future care models. Where the paper makes a prospective claim rather than reporting demonstrated benefit, that claim should be read as an explicit extrapolation from current evidence rather than as settled fact.

3 A framework for computational leverage

We propose five properties that determine whether a disease class is likely to benefit from larger inference budgets:

1. **Search-space size**: how many plausible hypotheses, targets, interventions, or sequences must be evaluated.
2. **Data burden**: how many modalities and how much longitudinal context are required to reason well.
3. **Personalization**: how much the correct answer depends on the specific patient rather than the average protocol.
4. **Actionability**: whether better synthesis can materially alter diagnosis, escalation, treatment choice, or trial eligibility.

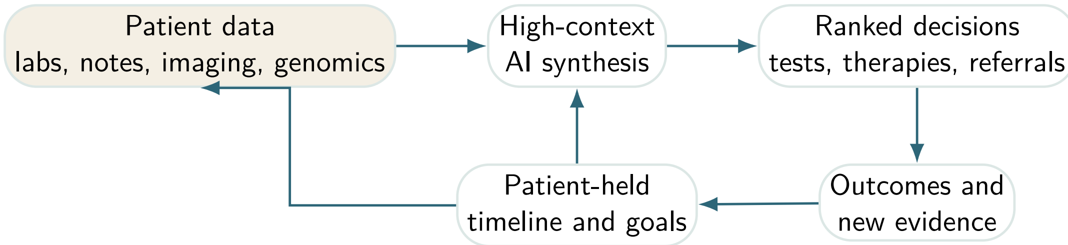

Figure 1: The patient-compute loop. In high-burden disease, value comes from repeated synthesis over the full timeline rather than one-shot question answering.

5. **Compute elasticity**: whether additional reasoning depth is likely to improve the result instead of merely restating the same answer more fluently.

Under this framework, precision oncology, rare-disease genomics, resistant infection, certain critical care syndromes, and selected long-horizon chronic conditions rise to the top. These domains combine large search spaces with heterogeneous data and decisions whose quality directly affects outcome. By contrast, diseases governed primarily by narrow protocols or limited action space may benefit more from automation than from massive inference.

This framework also clarifies when AI adds value beyond what a clinician can already provide. In domains with low compute elasticity, a clinician working at normal pace already reaches the correct decision—more thinking time, whether human or machine, buys nothing. In high-elasticity domains, every additional minute of deliberation can shift the differential, reorder treatment options, or surface a trial that would otherwise be missed. The practical reality is that clinicians cannot spend an hour reasoning through every case, even when an hour would help. AI does not replace clinical reasoning in these settings. It extends the reasoning budget available for the cases that need it most.

Crucially, while the scores in Table 1 are conceptual illustrations, they point toward a concrete, falsifiable research programme. These dimensions can be operationalized empirically: *search-space size* can be measured by counting the number of plausible differentials, approved therapies, or the relevant variant space for a typical case; *compute elasticity* can be quantified through experimental designs that systematically vary reasoning time or language-model inference depth, measuring the resulting change in downstream decision quality; and *data burden* can be assessed by counting the distinct data modalities routinely required before a defensible clinical decision can be made. Defining these metrics transforms the computational view from a descriptive metaphor into an actionable engineering target.

Why prevalence is not enough. Common diseases are not automatically the best targets for large inference budgets. The relevant question is whether more synthesis can change a meaningful patient-specific decision. That is why prevalence rankings and compute-opportunity rankings diverge.

4 Why patient-held data matter

The computational view becomes much stronger when we recognize that centralized medicine is already a distributed computation. The paper thus far largely frames reasoning as a single-agent problem—"a clinician" reasoning about a case. In reality, clinical reasoning is distributed across general practitioners, specialists, pathologists, radiologists, and multidisciplinary team meetings. Each referral is a computational step, and each handoff is a potential information loss. The "diagnostic odyssey" in rare disease—averaging

Domain	Search	Data	Pers.	Action	Elast.	Interpretation
Precision oncology	5	5	5	5	5	Dense multimodal evidence and many branching intervention paths make more inference directly decision-relevant.
Rare-disease genomics	5	5	5	4	5	Large hypothesis spaces and sparse distributed clues create unusually high value for long-context reasoning.
Resistant infection	4	4	4	5	4	The answer depends on organism, host state, prior exposure, and time-sensitive escalation.
Sepsis / ICU	4	5	4	5	4	Continuous streaming data create a high-burden monitoring problem in which timing dominates outcome.
Chronic inflammatory disease	3	4	4	4	3	Longitudinal symptom and biomarker integration can improve timing and sequencing, but action spaces are narrower.
Routine protocol care	1	2	1	2	1	Most value lies in reliable execution and logistics rather than larger inference budgets.

Table 1: Illustrative scoring matrix for computational leverage. Scores are conceptual rather than empirical, but they make the framework explicit and falsifiable.

7 years and 15 referrals to reach a diagnosis—is not a story about a single doctor who failed to think hard enough. It is the story of a distributed search that could not maintain state across its own nodes. Hypotheses get dropped at specialty boundaries, and prior exclusions are repeated because no single entity holds the full search history.

This reframing reveals what AI is uniquely positioned to do: not merely "think longer" in isolation, but hold the full search state together across the entire trajectory of fragmented care. This connects directly to the necessity of patient data access. The patient is the only entity that persists across all these episodes, making them the natural locus of computational state. Once the patient can assemble data across encounters, institutions, and modalities, compute can move with the patient rather than remaining trapped inside an institution. OpenNotes demonstrated that patients value direct access to clinicians' notes and that transparency can improve engagement with care [8]. Follow-on work showed that patients often shared those notes with family members and caregivers, turning the note into a coordination artifact rather than a static document [14]. In parallel, patient-generated data streams have become increasingly common across chronic disease, symptom tracking, wearables, and home monitoring [22]. The strategic implication is that records are no longer only institutional memory; they are becoming a substrate for patient-held, continuous reasoning.

Raw access, however, is not enough. Tuttle and colleagues argued that precision medicine will remain incomplete if patients are not active participants in how data are interpreted and used [30]. Hassan and colleagues similarly emphasize that AI for shared decision-making must improve understanding and deliberation, not simply generate recommendations [11]. Jim and colleagues reviewed the analytic and operational landscape of patient-generated health data in oncology and concluded that the integration challenge is not merely technical but also clinical, organizational, and evidentiary [15]. More recently, Salmi and colleagues showed in a proof-of-concept study that patients can already use large language models on open notes to interpret and work with their records [26]. That result should be read carefully: it is early evidence, not mature clinical validation. But it clarifies the direction of travel. Once patients can access timelines, test results, notes, and symptom logs, the missing layer is repeatable reasoning over that history.

4.1 Counterarguments and constraints

Patient-held data are not automatically clinically useful. Provenance may be uncertain, timestamps may be incomplete, self-tracked measures may drift in quality, and the patients most likely to benefit from longitudinal computation may also face the largest barriers in digital access. Huckvale and colleagues make a similar point for digital phenotyping: clinically interesting data streams do not become clinically usable unless quality, safety, equity, and implementation needs are addressed directly [12]. The computational paradigm therefore depends on a stricter standard than mere interoperability. It requires provenance, chronology, validation, and interfaces that separate observed evidence from inferred conclusions.

> **Access versus leverage.** Data access creates possibility. Computational interpretation creates leverage. In a patient-centered health stack, the two should be designed together.

5 Evidence from the highest-leverage domains

5.1 Precision oncology

Cancer is already a computational problem. Molecular diagnosis, resistance interpretation, treatment sequencing, and trial matching require synthesis across pathology, imaging, genomics, prior therapy, and time. Personalized cancer vaccines make that explicit: the workflow depends on identifying patient-specific tumor mutations, ranking neoantigens, and manufacturing a bespoke intervention. In a landmark study, Sahin and colleagues showed that individualized RNA mutanome vaccines could mobilize poly-specific immunity against melanoma [25]. Structure-prediction systems such as AlphaFold further expand the search surface by making protein consequence analysis more tractable across large molecular spaces [17].

The importance of compute here is not cosmetic. It sits inside the decision itself: which mutation matters, which escape mechanisms are plausible, which combination is worth pursuing, and which trial remains defensible for this patient now. Oncology is one of the clearest examples of a disease domain where the burden of reasoning has become part of the disease burden itself.

A concrete workflow makes the point clearer. The same patient may generate surgical pathology, radiology progression patterns, whole-exome or panel sequencing, circulating tumor DNA, treatment toxicities, and eligibility-relevant comorbidities. The computational task is to rank candidate drivers, resistance hypotheses, and therapeutic branches while making the uncertainty around each step visible. In other words, the leverage does not come from generating prose about cancer. It comes from compressing a large search problem into a smaller set of defensible next moves.

5.2 Rare disease genomics

Rare-disease diagnosis is where long-context reasoning becomes immediately clinically useful. The problem is not only finding a variant. It is aligning phenotype, pedigree, prior negative workup, and the literature fast enough to affect care. In 2024, Wojcik and colleagues reported in the *New England Journal of Medicine* that genome sequencing materially improved diagnosis for patients with suspected rare disease and supported its use as a first-tier test in appropriate settings [31].

This is a canonical compute-heavy domain: large hypothesis spaces, sparse signal, fragmented records, and a direct payoff when reasoning succeeds. The same logic applies to critically ill infants, undiagnosed neurologic disease, and multisystem pediatric cases where delay compounds. When diagnosis arrives late, the cost appears as additional procedures, avoidable admissions, and years spent navigating the wrong branch of care.

5.3 Resistant infection and antibiotic discovery

In antimicrobial resistance, the search problem has two layers. The first is patient-specific treatment selection: organism identity, susceptibility, host state, prior exposure, organ function, and drug interactions. The second is discovery itself: how to search chemical space for compounds with novel activity. Stokes and colleagues used deep learning to identify halicin, demonstrating that machine learning can materially accelerate antibiotic discovery in regions of chemical space that were not obvious to standard pipelines [28].

This is exactly the kind of problem where more compute can change outcomes: better regimen ranking in the clinic and better exploration of candidate space in the lab.

5.4 Sepsis and critical care

Critical care is different from oncology and rare disease, but it is no less computational. Here the challenge is time. Dozens of measurements update continuously, and the relevant signal is often distributed across trends rather than single observations. Adams and colleagues showed that a machine-learning sepsis early-warning system deployed prospectively across multiple hospitals was associated with faster antibiotic administration and lower mortality [2]. Rajkomar and colleagues had earlier shown that deep learning on electronic health records could scale across broad clinical prediction tasks using the raw longitudinal record itself [24]. Moses and colleagues extended this logic into pragmatic deterioration surveillance, showing that real-time systems can be attached to live escalation pathways [21].

These results matter because they establish two parts of the same claim: longitudinal records contain clinically actionable signal, and systems that can hold more of that context together can improve timing-sensitive decisions.

5.5 Psychiatry and mental health

While oncology, rare disease, and infection have relatively well-defined targets, psychiatry is arguably the most computationally complex domain in medicine. Mental health care involves enormous hypothesis spaces, longitudinal and multimodal data, and highly individual treatment responses (for example, first-line antidepressants fail in approximately 50% of cases). The current standard of care functions essentially as a brute-force search through medication options over months or years. Although objective biomarkers are often sparser than in oncology, the fundamental computational burden—iterative search under profound personal and clinical uncertainty—makes psychiatry a critical test case for any framework that treats medicine as a search and synthesis problem.

5.6 Patient-reported outcomes and symptom monitoring

Not all high-value computation depends on genomics. Basch and colleagues demonstrated that electronic patient-reported symptom monitoring during cancer treatment improved overall survival, likely by enabling earlier clinical response to deterioration patterns that would otherwise be missed [4, 5]. This is an important bridge between institutional and patient-held data. Symptoms, side effects, and treatment tolerability often live outside the hospital until they are severe enough to trigger contact. In a computational paradigm, those signals become first-class inputs into the case model.

6 N-of-1 medicine and personal computation

The computational view converges naturally with N-of-1 medicine. Schork argued that precision medicine should increasingly be understood at the level of the individual rather than the cohort average [27]. That argument has a formal clinical-trial lineage. Guyatt and colleagues laid much of the early foundation for randomized trials in individual patients [10], and Lillie and colleagues later argued that N-of-1 trials are unusually well aligned with the actual endpoint of clinical practice: optimizing care for a single person [18]. More recent work has pushed this further. Augustine and colleagues proposed a roadmap for N-of-1 therapies, describing the scientific, operational, and regulatory requirements for highly individualized interventions [3]. Jonker and colleagues similarly outline how N-of-1 medicine could reshape the development and evaluation of treatments for rare and heterogeneous disease [16]. These papers matter because they move the discussion from personalization as a slogan to personalization as an organizing scientific unit.

Longitudinal self-measurement provides a parallel but distinct line of evidence. Chen and colleagues showed in 2012 that integrative personal omics profiling could reveal dynamic molecular and medical phenotypes within a single individual over time [6]. Piening and colleagues later demonstrated that personal, dense, dynamic data clouds can be used to monitor health transitions with much greater temporal resolution than standard episodic care [23]. These studies do not imply that every patient should become a self-quantification project. They do show that biology, symptoms, and risk are often more legible in longitudinal within-person data than in sparse snapshot encounters.

Public self-experimentation should be interpreted separately again. Highly instrumented self-tracking programmes such as Bryan Johnson's are not clinical evidence and should not be treated as such. What they do reveal is latent demand: some people already want to run dense longitudinal data loops over their own bodies. The scientific path forward is not celebrity optimization. It is the translation of that demand into audited N-of-1 workflows that combine patient goals, validated measurements, and clinically accountable reasoning.

7 Case examples

Table 2 shows why a single computational model fits these otherwise different domains.

7.1 Concrete pathways for future use

Figure 2 makes the future-oriented claim concrete. In each case, the point of computation is not to replace the clinician. It is to improve preparation, ranking, and timing for the next branch in the care pathway.

1. **Precision oncology pathway.** A patient-assembled case file combines sequencing, pathology, imaging, and prior therapies. Compute is used to rank candidate drivers, plausible resistance mechanisms, toxic trade-offs, and live trials. The output is not "the answer" but a defensible next branch with explicit uncertainty.

Domain	Core data	How computation is leveraged	Representative evidence
Precision oncology	Genomics, pathology, imaging, treatment history	Variant interpretation, resistance modeling, neoantigen ranking, regimen sequencing, trial matching	Sahin et al. [25]; Jumper et al. [17]
Rare disease	Phenotype timeline, pedigree, genome sequencing, prior workup	Cross-linking phenotype ontologies, literature, family history, and candidate variants	Wojcik et al. [31]
Resistant infection	Microbiology, susceptibilities, organ function, prior antibiotic exposure	Better ranking of therapies and faster search across compound or regimen space	Stokes et al. [28]
Sepsis / ICU	Streaming vitals, labs, notes, medication events	Earlier escalation from weak, time-distributed patterns across the longitudinal record	Adams et al. [2]; Rajkomar et al. [24]; Moses et al. [21]
Patient-reported symptoms	Patient-reported outcomes, treatments, toxicities, goals	Detecting deterioration and treatment intolerance before they become acute	Basch et al. [4]; Basch et al. [5]
Patient-held records	Notes, meds, labs, imaging reports, symptoms, goals	Converts raw data access into usable reasoning leverage before and between encounters	Delbanco et al. [8]; Salmi et al. [26]

Table 2: High-value disease areas under a computational lens.

2. **Rare disease pathway.** Parents or clinicians maintain a phenotype timeline with milestones, regressions, imaging, and prior exclusions. Compute links that timeline to genome evidence and the literature, reducing time spent re-deriving the case from scratch at each tertiary referral.

3. **Chronic inflammatory pathway.** A patient with a relapsing condition integrates symptoms, labs, wearable data, and medication exposure over time. Compute is used within-person, not only across populations, to identify recurrent precursors of flare, intolerance, or non-response.

8 Evaluation and endpoints

If the computational view is correct, it should change how health AI is evaluated. Average discrimination metrics and point-estimate accuracy are not enough. In high-stakes medicine, calibration matters as much as accuracy. Measuring how well a system quantifies its own uncertainty can be more valuable than merely evaluating its top-choice correctness: a system that outputs "60% diagnosis A, 25% diagnosis B, and here is the test that discriminates between them" is radically more useful than a bare point estimate of A, even at identical baseline accuracy. This connects directly to compute elasticity. Sometimes more reasoning does not change the top-ranked answer but dramatically sharpens the uncertainty estimate around it, which clinically determines whether a clinician will order a confirmatory test, refer to a specialist, or treat empirically. The computational view must therefore care about decision-relevant uncertainty at least as much as it cares

Future patient personas

Patient	Data substrate	Computational leverage
Asha, 58 metastatic NSCLC	Tumour DNA/RNA, pathology, CT trends, prior lines, toxicities, performance status	Variant prioritization, resistance hypotheses, neoantigen ranking, and live trial matching change the next therapeutic branch.
Mateo, 6 undiagnosed neurodevelopmental syndrome	Phenotype timeline, EEG, MRI, family history, genome sequencing, prior negative workup	Phenotype-variant ranking and literature synthesis narrow candidate mechanisms and guide targeted confirmatory testing.
Leila, 34 Crohn's disease on biologics	Symptoms, calprotectin, medication adherence, wearables, sleep, diet, clinician notes	Within-person pattern detection turns fragmented self-tracking into actionable flare prediction and better-timed escalation.

Figure 2: Concrete personas for a computational health stack. The same design principle appears across specialties: assemble the full timeline, compute over it repeatedly, and use the result to improve the next decision.

about the prediction itself. In the domains highlighted here, the relevant question is whether synthesis quality—including calibration—improved at the point where the case could still change direction.

Illustrative translational case
A widely discussed 2025 media report described how an Australian entrepreneur used sequencing, variant analysis, and AI-assisted reasoning to help design a personalized mRNA treatment strategy for his dog Rosie, who had a mast cell tumour [1]. This is not high-level evidence in the way a randomized trial is. It is useful for a different reason: it makes visible what the computational workflow looks like once motivated people can assemble data, models, and synthesis outside a traditional institutional pipeline.

9 What should be built

The implication is not that every patient should be left alone with a chatbot. The implication is that health systems, startups, and regulators should prioritize tools that make high-burden reasoning portable, inspectable, and patient-aligned.

1. **Patient-assembled case files.** Systems should ingest records from multiple providers and preserve chronology rather than flattening everything into isolated visits.
2. **Explicit uncertainty handling.** Models should separate evidence, inference, and missing information, especially in oncology, genetics, psychiatry, and infection.
3. **Decision-oriented outputs.** The output should not just summarize. It should rank the next tests, consultations, or therapeutic branches.
4. **Re-runnable reasoning.** Every major disease course changes over time. The right unit is a living case model, not a one-time answer.

Study type	What changed	Why it matters for the thesis
Pragmatic randomized clinical trial in cardiovascular screening	AI-enabled ECG alerts were tested as an intervention rather than a retrospective classifier [19]	Supports the claim that computational systems can change downstream decisions and outcomes, not merely generate scores.
Pragmatic cluster-randomized deterioration trial	Real-time surveillance for patient deterioration was tested across live inpatient settings [21]	Strengthens the case that high-context clinical surveillance can function as an operational care system.
Patient-record sharing study	Patients with transparent notes shared them with family and caregivers, extending the practical reach of the record [14]	Shows that once patients have access to records, those records begin acting as coordination infrastructure.
Patient-side LLM proof of concept	Patients used large language models on open notes to better understand and work with their records [26]	Gives direct support to the argument that personal compute can become part of the patient toolkit.

Table 3: Evidence that the computational view extends beyond retrospective prediction.

5. **Patient-clinician collaboration.** The design target is a stronger clinical conversation, not consumer isolation.

6. **N-of-1 evidence layers.** Systems should support repeated within-person hypothesis testing when disease is relapsing, heterogeneous, or highly treatment-sensitive.

10 Risks and constraints

The computational view does not weaken the need for evidence or clinical oversight. It sharpens it. High-burden disease is precisely where hallucinated synthesis, hidden bias, dataset shift, and brittle prompting are most dangerous. Better reasoning systems must therefore be auditable, source-grounded, and explicit about what they do not know.

There is also a distributional risk: if compute meaningfully improves outcomes, unequal access to compute will become a health inequity. Data access without reasoning access will be incomplete empowerment. A future health stack that gives patients records but not useful synthesis will still leave too much leverage with the institution.

The paper also argues against a common failure mode in AI health discourse: treating "personalization" as a vague user-experience feature. In the computational paradigm, personalization is a property of the reasoning problem itself. It requires timeline integrity, provenance, repeated re-analysis, and outcome feedback.

11 Operational consequences

If this view is correct, the agenda for health AI changes. Product teams should stop treating every workflow as though it deserves the same level of model complexity. Clinical systems should separate protocol problems from search problems. Regulators should ask not only whether a model is accurate on average, but whether it improves the quality of reasoning in the cases where reasoning is the bottleneck. And patient-facing tools

Domain	Primary endpoint candidates	Why these endpoints fit the computational thesis
Rare disease	Time to diagnosis; diagnostic yield; avoided procedures	Measures whether synthesis reduced time spent on the wrong branch of care.
Precision oncology	Time to molecular recommendation; trial-match yield; change in treatment ranking; time to next line	Measures whether computation improved branching decisions rather than documentation alone.
Sepsis / ICU	Time to escalation; antibiotics or intervention timing; false alerts per patient-day; mortality or ICU transfer	Measures whether longitudinal synthesis improved timing without overwhelming workflow.
Chronic disease / N-of-1 care	Time to flare detection; within-person treatment response; symptom burden; adherence to preferred regimen	Measures whether repeated computation improves individualized management over time.
Patient-facing record synthesis	Patient comprehension; visit preparedness; question quality; fewer duplicated tests; continuity across sites	Measures whether access plus reasoning improved the practical utility of records.

Table 4: Evaluation endpoints aligned with a computational view of medicine. The goal is to measure whether reasoning improves decisions, not only whether predictions look accurate offline.

should be built around timeline integrity, evidence provenance, and re-runnable case synthesis rather than one-shot conversational convenience.

The practical payoff is concentration. More compute should go to the places where it buys something real: fewer missed driver mutations, faster rare-disease diagnoses, better antibiotic choices, earlier sepsis escalation, more systematic medication sequencing in psychiatry, cleaner preparation for specialist visits, and better continuity when care fragments across institutions. Results from pragmatic intervention studies in ECG alerting, patient deterioration surveillance, and electronic symptom monitoring suggest that this shift can move from theory into deployed operations when the system is attached to a real decision path [4, 19, 21].

Testable predictions

Prediction 1. Within five years, the most effective clinical AI deployments will increasingly be evaluated on time-to-decision and branch-quality endpoints rather than stand-alone discrimination metrics.

Prediction 2. The first widely adopted patient-facing AI tools in medicine will succeed not as synthetic clinicians but as timeline integrators and case-preparation systems.

Prediction 3. N-of-1 computational care loops will emerge first in rare disease, oncology, and relapsing chronic disease, where within-person variation and high decision burden are most visible.

Prediction 4. Institutions that expose records without exposing usable reasoning will create a new form of incomplete patient empowerment.

12 Conclusion

The next decade of medical AI should be judged less by how often it automates paperwork and more by how often it helps solve the hard cases. The diseases that matter most here are not simply the most prevalent ones. They are the ones that become search problems when viewed at patient resolution. Precision oncology, rare-disease genomics, resistant infection, selected critical care settings, and some chronic relapsing diseases are early proof that this lens is practical rather than rhetorical.

The broader shift is cultural as much as technical. Once patients can assemble their own records and run high-context reasoning over them with clinicians, compute becomes part of care capacity itself. That is why a computational view of medicine matters: it provides a principled way to direct AI toward settings where additional inference can genuinely change outcomes, and where health understanding can become a shared, longitudinal capability rather than a privilege of the institution.

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